

Problem 1 : Hölder Continuous Functions

Let $f : [0, 1] \rightarrow \mathbb{R}$. We say that f is Hölder continuous of order $\alpha \in (0, 1)$ and write $f \in C^\alpha([0, 1])$ if

$$\|f\|_{C^\alpha} := \sup\{|f(x)|; x \in [0, 1]\} + \sup\left\{\frac{|f(x) - f(y)|}{|x - y|^\alpha}; x, y \in [0, 1], x \neq y\right\} < \infty.$$

For $f, g \in C^\alpha([0, 1])$, define $d(f, g) = \|f - g\|_{C^\alpha}$.

(a) Show that $(C^\alpha([0, 1]), d)$ is a complete metric space.

(b) Let (f_n) be a sequence in $C^{1/2}([0, 1])$ such that $\|f_n\|_{C^{1/2}} \leq 1$ for $n = 1, 2, \dots$. Show that (f_n) has a subsequence that converges in $C^{1/3}([0, 1])$.

Solution.

Problem 2 : Connected Subsets Covering a Connected Space

Let X be a connected metric space and let $A, B \subseteq X$ be closed subsets such that $X = A \cup B$. Assume also that $A \cap B$ is connected. Show that A and B are connected.

Solution. Assume the contrary that A or B are disconnected. WLOG let A be disconnected. Thus $\exists U, V \in \mathcal{T}(X)$ such that $A \cap (U \cup V) = A$ and $U \cap V = \emptyset$.

Problem 3 : Connectedness and Completeness of a Lipschitz Function Space

Let $C([0, 1])$ denote the space of continuous real-valued functions on $[0, 1]$ equipped with the uniform metric. Let X be a subset of $C([0, 1])$ defined via

$$X = \{f \in C([0, 1]); f(0) = 0, |f(x) - f(y)| \leq |x - y|\}.$$

Show that X is connected and complete.

Solution.

Problem 4 : Epsilon-Chain Connectedness Implies Connectedness

Let $K \subset \mathbb{R}^n$ be compact. Suppose that for every $\varepsilon > 0$ and for every pair $a, b \in K$ there is an integer $n \geq 1$ and a sequence of points $x_0, \dots, x_n \in K$ such that

$$x_0 = a, \quad x_n = b, \quad |x_k - x_{k-1}| \leq \varepsilon, \quad 1 \leq k \leq n.$$

Show that K is connected.

Hint. You may wish to prove the following general fact first: let (X, d) be a metric space and assume that $F \subseteq X$ is closed and $K \subseteq X$ is compact. Show that if F and K are disjoint, then

$$\inf_{x \in F, y \in K} d(x, y) > 0.$$

Solution.